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Physics 180 THV-1
Problem Set 3

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Problem 3.1

Convince yourself (and your parents and your friends and your dog and most importantly your Physics 180 professor as well as the checker Mr. Kong) that the total degeneracy of states in the harmonic oscillator potential case is indeed given by

$$(\Lambda + 1)(\Lambda + 2) \tag{1.1}$$

Solution

Proof. From [1], we have the definition of Λ in terms of the principal and angular quantum numbers.

$$\Lambda = 2n + \ell - 2 \tag{1.2}$$

and the energy eigenvalues

$$E_{nl} = \hbar\omega\left(\Lambda + \frac{3}{2}\right) \tag{1.3}$$

From this, we can infer that every combination of n and ℓ producing the same Λ have the same energy.

We can solve for ℓ from Λ

$$\ell = \Lambda - 2(n - 1) \tag{1.4}$$

and since $n = 1, 2, 3, \dots$, the allowed values for ℓ for a given Λ are

$$\ell = \Lambda, \Lambda - 2, \Lambda - 4, \dots$$

With the sequence terminating at $\ell = 0$ for an even Λ , and $\ell = 1$ for odd Λ .

For each ℓ , there are $2\ell + 1$ possible values of the magnetic quantum number m , while the factor of 2 accounts for the two possible spin states.

Again, given from [1], we are given the number of states with degenerate energy levels as:¹

$$n_d = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) \tag{1.5}$$

However, since the value of ℓ is constrained, we have

$$n_d = 2 \sum_{\ell=\Lambda, \Lambda-2, \dots} (2\ell + 1) \tag{1.6}$$

and this might have different results depending on even or odd Λ . We evaluate this separately for both cases

¹This was proven in Quiz 3

Even Λ

Let $\Lambda = 2k$. Then, the allowed values of ℓ are

$$\ell = 2k, 2k - 2, \dots, 2, 0 \quad (1.7)$$

Thus, we have the sum:

$$\begin{aligned} \sum_{\ell=\Lambda, \Lambda-2, \dots} (2\ell + 1) &= (2(2k) + 1) + (2(2k - 2) + 1) + \dots + (2(2 - 2) + 1) \\ &= (4k + 1) + (4k - 3) + \dots + 5 + 1 \end{aligned}$$

This is an arithmetic series, with $k + 1$ terms (from k to 0 inclusive). Hence, the arithmetic series is

$$\begin{aligned} \sum_{\ell=\Lambda, \Lambda-2, \dots}^{\text{even}} (2\ell + 1) &= (k + 1) \frac{(4k + 1) + 1}{2} \\ &= \frac{(k + 1)(4k + 1) + k + 1}{2} \\ &= \frac{4k^2 + 5k + 1 + k + 1}{2} \\ &= \frac{4k^2 + 6k + 2}{2} \\ &= 2k^2 + 3k + 1 \\ &= (k + 1)(2k + 1) \end{aligned}$$

Since $\Lambda = 2k$, then

$$\begin{aligned} \sum_{\ell=\Lambda, \Lambda-2, \dots}^{\text{even}} (2\ell + 1) &= \left(\frac{\Lambda}{2} + 1 \right) (\Lambda + 1) \\ &= \left(\frac{\Lambda + 2}{2} \right) (\Lambda + 1) \\ &= \frac{(\Lambda + 2)(\Lambda + 1)}{2} \end{aligned}$$

Odd Λ

Let $\Lambda = 2k + 1$. Then

$$\ell = 2k + 1, 2k - 1, \dots, 3, 1$$

Therefore, the series is

$$\begin{aligned} \sum_{\ell=\Lambda, \Lambda-2, \dots}^{\text{odd}} (2\ell + 1) &= (2(2k + 1) + 1) + (2(2k - 1) + 1) + \dots + 7 + 3 \\ &= (4k + 3) + (4k - 1) + \dots + 7 + 3 \end{aligned}$$

Once again, this has $k + 1$ terms, from 0 to k inclusive. Therefore, the total sum of the

arithmetic series is

$$\begin{aligned}
\sum_{\ell=\Lambda, \Lambda-2, \dots}^{\text{odd}} (2\ell + 1) &= (k+1) \frac{(4k+3) + 3}{2} \\
&= \frac{4k^2 + 7k + 3 + 3k + 3}{2} \\
&= \frac{4k^2 + 10k + 6}{2} \\
&= 2k^2 + 5k + 3 \\
&= (k+1)(2k+3)
\end{aligned}$$

However, $\Lambda + 1 = 2k + 2$, and $\Lambda + 2 = 2k + 3$. Hence

$$\sum_{\ell=\Lambda, \Lambda-2, \dots}^{\text{odd}} (2\ell + 1) = \frac{(\Lambda + 1)(\Lambda + 2)}{2} \quad (1.8)$$

Thus, for both cases, we have

$$\sum_{\ell=\Lambda, \Lambda-2, \dots} (2\ell + 1) = \frac{(\Lambda + 1)(\Lambda + 2)}{2} \quad (1.9)$$

Since (1.6)

$$n_d = 2 \sum_{\ell=\Lambda, \Lambda-2, \dots} (2\ell + 1) \quad (1.10)$$

then

$$n_d = 2 \cdot \frac{(\Lambda + 1)(\Lambda + 2)}{2} \quad (1.11)$$

Therefore:

$$\boxed{n_d = (\Lambda + 1)(\Lambda + 2)} \quad (1.12)$$

□

Thank you, Mr. Kong.

References

- [1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, [UvLé](#).

Appendices

Code

No code was used for this problem set